

Component Order List – Electrochemical Impedance Spectroscopy PCB

Cross-checked against the attached Project-1.xlsx BOM export: no differences found between the BOM and the actual schematic/PCB component list. Preferred supplier: reichelt elektronik (Germany). Where a part is not stocked at Reichelt, an alternative German-accessible distributor is listed instead.

Component	Qty	Manufacturer Part Number	Order Link	Notes
Crystal, 8 MHz, SMD (5x3.2mm)	1	ABM3B-8.000MHZ-B2-T (Abracon)	Digikey.de	Not stocked at Reichelt
Crystal, 25 MHz, SMD (3.2x2.5mm)	1	ASE-25.000MHZ-LC-T (Abracon)	Farnell.de	Not stocked at Reichelt
Capacitor, 1 nF, THT axial, 7.5mm pitch	1	generic film/ceramic 1nF	Reichelt (search)	Exact product page not individually confirmed
Capacitor, 100 nF, THT axial, 7.5mm pitch	13	WIMA MKS4, 100nF, 100V, RM7.5	Reichelt #41124	Confirmed match
Capacitor, 3.3 nF, THT axial, 5mm pitch	1	FKS2-100, 3.3nF, 100V, RM5	Reichelt #172543	Confirmed match
Capacitor, 1 uF, THT axial electrolytic	4	Vishay, 1uF, 63V, axial	Reichelt #230299	Confirmed match
Capacitor, 22 pF, THT ceramic	2	FREI, 22pF, NPO, 100V, RM2.54	Reichelt #9281	Lead spacing 2.54mm vs 7.5mm on board - re-bend leads
Capacitor, 10 uF, THT axial electrolytic ("cup")	2	generic axial electrolytic 10uF	Reichelt (category)	Exact product page not individually confirmed
Connector, JST XH, 5-pin, THT, 2.5mm pitch	1	B5B-XH-A(LF)(SN) (JST)	Digikey.de	Not confirmed at Reichelt
LDO voltage regulator, TO-92, 3.3V	1	MCP1700T-3302E/TO (Microchip)	Digikey.de	Confirm exact voltage grade against schematic; not confirmed at Reichelt
Op-amp, dual, precision JFET, SOIC-8	2	OPA2140AID (Texas Instruments)	Digikey.de	Not stocked at Reichelt
Resistor, 10 kΩ, THT axial, 1/4W	6	0207 carbon film, 5%	Reichelt #1338	Confirmed match
Resistor, 5.1 kΩ, THT axial, 1/4W	2	0207 carbon film, 5%	Reichelt #1436	Confirmed match
Resistor, 1 kΩ, THT axial, 1/4W	1	0207 carbon film, 5%	Reichelt #1315	Confirmed match

Component	Qty	Manufacturer Part Number	Order Link	Notes
Resistor, 22 Ω , THT axial, 1/4W	2	0207 carbon film, 5%	Reichelt #1381	Confirmed match
Resistor, 4.7 k Ω , THT axial, 1/4W	1	0207 carbon film, 5%	Reichelt #1425	Confirmed match
Resistor, 200 Ω , THT axial, 1/4W	1	0207 carbon film, 5%	Reichelt (search)	Exact product page not individually confirmed
DDS signal generator, MSOP-10	1	AD9833BRMZ (Analog Devices)	Reichelt #112110	Confirmed match, EUR 18.15
SPI Flash, 32 Mbit, SOIC-8	1	W25Q32JVSSIQ (Winbond)	Digikey.de	Not stocked at Reichelt
SWD connector, 2x5, 1.27mm pitch, box header	1	generic 10-pin 1.27mm header	Reichelt #262741	This is a straight (not shrouded/keyed) header - verify against schematic footprint
Test points (bare copper pad)	19	n/a - no physical part	n/a	Bare pads only; order turret/testpoint pins separately if physical posts are wanted
Microcontroller, STM32F303CCT6, LQFP-48	1	STM32F303CCT6 (STMicroelectronics)	Reichelt #265379	Confirmed match
USB-C receptacle, SMD, 12-pin	1	TYPE-C-31-M-12 (Hroparts Elec)	LCSC.com	Not stocked at Reichelt
USB-UART bridge, CP2102, QFN-28	1	CP2102-GM (Silicon Labs)	Reichelt #353966	Confirmed match, EUR 8.99
Voltage reference, precision, SOIC-8	1	ADR4533BRZ (Analog Devices)	Digikey.de	Not stocked at Reichelt